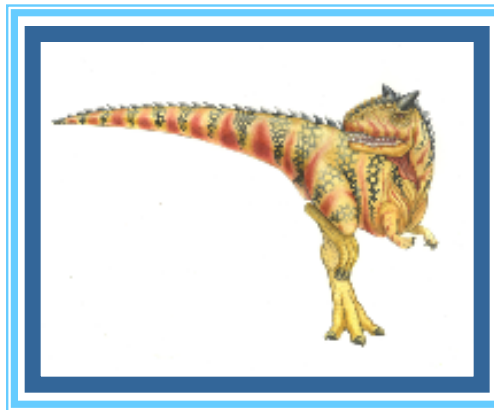


File-System Interface





Outline

- File Concept





File Concept

- Contiguous logical address space
- Types:
 - Data
 - 4 Numeric
 - 4 Character
 - 4 Binary
 - Program
- Contents defined by file's creator
 - Many types
 - 4 **text file,**
 - 4 **source file,**
 - 4 **executable file**





File Attributes

- **Name** – only information kept in human-readable form
- **Identifier** – unique tag (number) identifies file within file system
- **Type** – needed for systems that support different types
- **Location** – pointer to file location on device
- **Size** – current file size
- **Protection** – controls who can do reading, writing, executing
- **Time, date, and user identification** – data for protection, security, and usage monitoring
- Information about files are kept in the directory structure, which is maintained on the disk
- Many variations, including extended file attributes such as file checksum
- Information kept in the directory structure





File Operations

- **Create**
- **Write** – at **write pointer** location
- **Read** – at **read pointer** location
- **Reposition within file** - **seek**
- **Delete**
- **Truncate**
- **Open (F_i)** – search the directory structure on disk for entry F_i , and move the content of entry to memory
- **Close (F_i)** – move the content of entry F_i in memory to directory structure on disk



THANK YOU

